

Question	Marking Guidance	Mark	Comments
07.1	Experiment 1 with 3 bands and experiment 2 with 5 bands in correct positions = 2 marks;; Experiment 1 with 3 bands and experiment 2 with 5 bands but not in correct positions = 1 mark; Experiment 1 Experiment 2 	2 (2 x AO2)	For two marks, the position of the second band from the top must be in the same position for both experiments. The lengths of the bands do not need to be the same, but the relative positions should be similar to the diagram shown.

Question	Marking Guidance	Mark	Comments
07.2	1. (Separate) DNA fragments/ladder of known sizes/lengths; 2. Compare position/distance/bands with unknown fragment(s);	2 (2 x AO2)	1. Ignore mass

Question	Marking Guidance	Mark	Comments
07.3	4 or four;	1 (1 x AO2)	

Question	Marking Guidance	Mark	Comments
07.4	1. Restriction endonucleases/enzymes cuts <u>plasmid</u> OR Restriction endonucleases/enzymes produce 'sticky ends'; 2. Ligase joins gene/DNA and plasmid OR Ligase joins 'sticky ends' OR Ligase forms phosphodiester bonds;	2 (2 x AO1)	1. Accept 'cuts DNA of <u>plasmid</u>'. 1. Ignore restriction enzymes cuts out the gene. 2. Ignore reference to hydrogen bonds or joining complementary base pairs.

